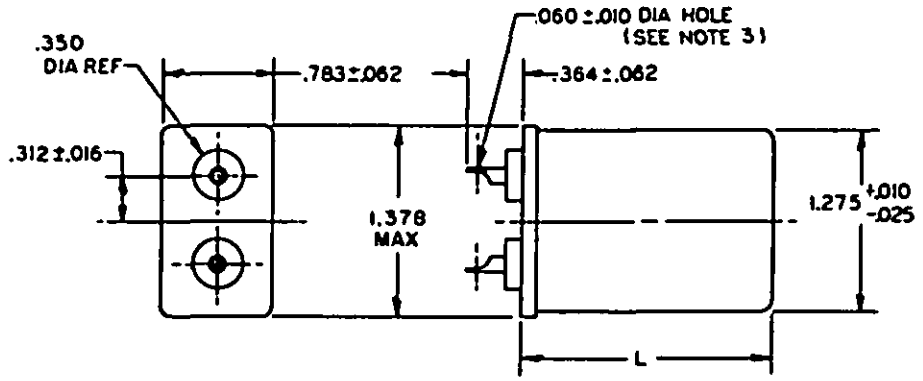
MILITARY SPECIFICATION SHEET

CAPACITORS, FIXED, ELECTROLYTIC (NONSOLID ELECTROLYTE), TANTALUM, (NONPOLARIZED, ETCHED FOIL), 85°C (VOLTAGE DERATED TO 125°C), STYLE CL54

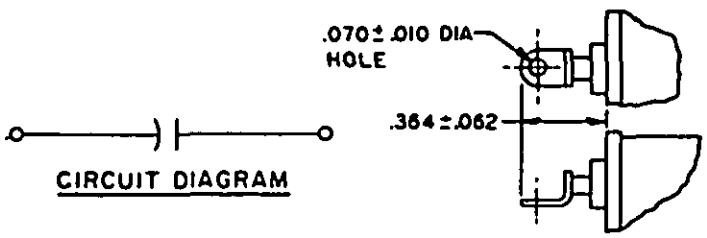
INACTIVE FOR NEW EQUIPMENT DESIGN. NO SUPERSEDING DOCUMENT.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the capacitors described herein shall consist of this document and the latest issue of Specification MIL-C-3965.



Dimensions	
	L ±.062 (1.57)
D1	1.375 (34.93)
D2	2.000 (50.80)
D3	2.500 (63.50)



CIRCUIT DIAGRAM

OPTIONAL ALTERNATE TERMINAL CONSTRUCTION

INCHES	MM	INCHES	MM
.010	.25	.312	7.92
.016	.41	.350	8.89
.025	.64	.364	9.25
.060	1.52	.783	19.89
.062	1.57	1.275	32.39
.070	1.78	1.378	35.00

NOTES:
1. Dimensions are in inches.
2. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
3. Orientation of terminal hole not fixed.

Ⓗ FIGURE 1. Style CL54 capacitors.

Ⓗ denotes changes

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T03:00Z
Check the source to verify that this is the current version before use.

FSC 5910

TABLE I. Capacitor characteristics.

Type designation	Voltage			Capacitance	DC leakage (max)			Impedance (max)	Case size
	Rated (85°C)	Derated (125°C)	Surge (85°C)		25°C	85°C	125°C		
	volts, dc	volts, dc	volts, dc	µF	µA	µA	µA	ohms	
CL54BE751UNG	15	10	17.2	750	52	400	500	11	D1
CL54BE152UNG	15	10	17.2	1,500	105	840	1,050	6	D2
CL54BE212UNG	15	10	17.2	2,100	147	1,175	1,470	4	D3
CL54BH721UNG	30	20	34.5	720	100	800	1,000	12	D2
CL54BH102UNG	30	20	34.5	1,000	140	1,120	1,400	8	D3
CL54BJ201UNG	50	30	57.5	200	42	335	420	20	D1
CL54BJ401UNG	50	30	57.5	400	84	670	840	10	D2
CL54BJ551UNG	50	30	57.5	550	115	920	1,150	7.2	D3
CL54BL201UNG	75	50	86.2	200	70	560	700	20	D2
CL54BL271UNG	75	50	86.2	270	94	750	940	15	D3
CL54BI121UNG	112	75	129.0	120	63	505	630	33	D2
CL54BI161UNG	112	75	129.0	160	84	670	840	25	D3
CL54BQ101UNG	150	100	172.0	100	70	560	700	40	D3

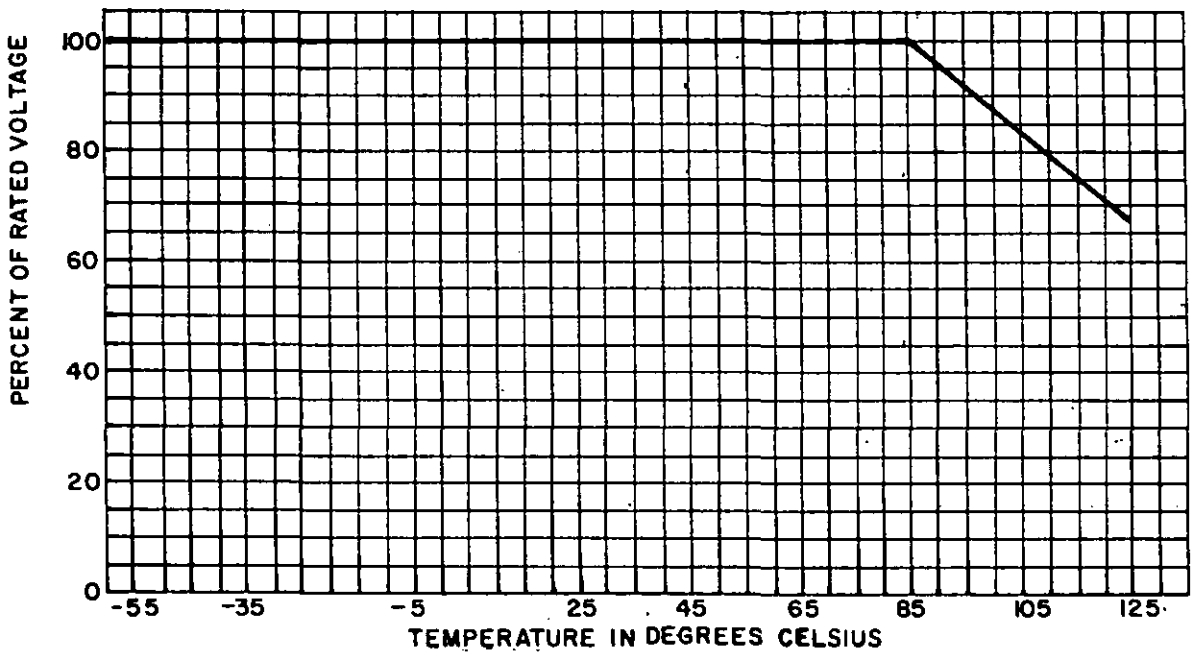


FIGURE 2. Voltage derating with temperature.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T03:00Z
Check the source to verify that this is the current version before use.

REQUIREMENTS:

Design and construction:
Dimensions and configuration:
Capacitors - See figure 1.
Brackets - See figure 3.
Case type - Rectangular, uninsulated.
Seal type - Hermetic (symbol G).
Terminals - See figure 1.
Rated temperature - -55° C to +85° C, voltage derated to 125° C (see figure 2).
Rated voltage - See table I.
Capacitance (cap) (nom) - See table I.
Capacitance tolerance - Symbol U (+75, -15%).

DC leakage (DCL) (max) (at +25° C):
See table I.

Dissipation factor (DF) (max): 30% (15 v), 20% (30 thru 75 v), or 15% (112 and 150 v).

Seal: In accordance with MIL-C-3965.

Dielectric withstanding voltage (DWV):
Sleeving to case (min) - Not applicable.
Terminal to case - Not applicable.

Insulation resistance:
Sleeving to case (min): Not applicable.
Terminal to case - Not applicable.

Temperature cycling:
DCL (max) - 125% of 25° C value of table I.
ΔCap - Within ±5% of initial measurement.
DF (max) - 115% of initial requirement.

Solderability: Not applicable.

Terminal strength:
Pull test - In accordance with MIL-C-3965.
Bend test (if req.) - In accordance with MIL-C-3965.
Torque test - Not applicable.
Wire-lead bend test - Not applicable.
Twist test - Not applicable.

Surge voltage: See table I.

Moisture resistance:
DCL (max) - 125% of 25° C value of table I.
ΔCap - Within ±5% of initial measurement.
DF (Max) - 115% of initial requirement.

Low temperature (storage):
DCL (max) - See table I (25° C).
ΔCap - Within ±5% of initial measurement.
DF (max) - Shall not exceed initial requirement.

Stability at low and high temperature:
Step 1 (+25° C):
DCL (max) - See table I.
Cap - Within tolerance of table I value.
DF (max) - Shall not exceed initial requirement.
Step 2 (-55° C):
Impedance - See table I.
ΔCap (max) - -65% (≤30 v) or -45% (>30 v) of step 1 value.
Step 3 (+25° C):
DCL (max) - See table I.
ΔCap - Within ±5% of step 1 value.
DF (max) - Shall not exceed initial requirement.
Step 4 (+85° C):
DCL (max) - See table I.
ΔCap (max) - +75% of step 1 value.
DF (max) - 50%.
Step 5 (+125° C):
DCL (max) - See table I.
ΔCap (max) - +75% of step 1 value.
DF (max) - 50%.
Step 6 (+25° C):
DCL (max) - See table I.
ΔCap - Within ±5% of initial measurement.
DF (max) - Shall not exceed initial requirement.

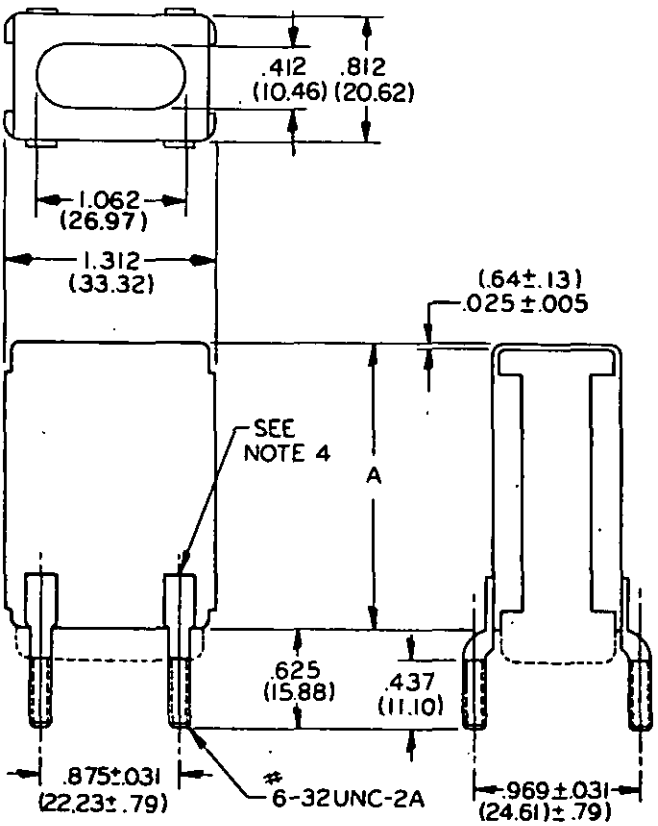
Reverse voltage: Not applicable.

Life at +85° C:
2,000-hr (qualification):
DCL (max) - See table I (25° C).
ΔCap - Within +15, -20% of initial measurement.
DF (max) - 130% of initial requirement.
240-hr (group B):
DCL (max) - See table I (25° C).
ΔCap - Within ±10% of initial measurement.
DF (max) - 130% of initial requirement.
1,760-hr (group C):
DCL (max) - See table I (25° C).
ΔCap - Within +15, -20% of initial measurement.
DF (max) - 130% of initial requirement.

Barometric pressure: Rated voltage.

Life at +125° C:
DCL (max) - See table I (25° C).
ΔCap - Within ±25% of initial measurement.
DF (max) - 150% of initial requirement.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T03:00Z
Check the source to verify that this is the current version before use.



BRACKETS FOR TYPE D CASE

Type designation	Case size	Dimensions						
		Dim	Inches			Millimeters		
			Min	Nom	Max	Min	Max	
CL050SD13	D1	A	1.159	1.175	1.191	29.44	30.25	
CL050SD23	D2	A	1.784	1.800	1.816	45.31	46.13	
CL050SD33	D3	A	2.284	2.300	2.316	58.01	58.82	

NOTES:
1. Dimensions are in inches.
2. Millimeters are in parentheses.
3. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
4. Spade lugs shall be rigid (either double welded, double riveted, single riveted and soldered, or single riveted and welded to the bracket).
5. Unless otherwise specified, tolerance is ±.062 (1.57 mm).

FIGURE 3. CL050 bracket.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T03:00Z
Check the source to verify that this is the current version before use.

Marking: Capacitors shall be marked as shown in the following example:
Example:
CL54BE751UNG - Type designation.
12345 7015 - Source and date code.
750 μF 15 vdc - Capacitance and rated voltage (at 85° C).

Voltage groups (85° C rated voltage):
I - - - - - - - - - - - - - 15 to 75 v, incl.
II - - - - - - - - - - - - - 112 and 150 v.

Substitution data. Capacitors covered by this specification are directly substitutable for capacitors of equivalent capacitance, capacitance tolerance, and voltage in the same case size.

Custodians:
Army - EL
Navy - EC
Air Force - 85

Review activities:
Army - MU
Navy - SH
Air Force - 11, 17, 80
DSA - ES

User activities:
Army -
Navy - AS, MC, OS
Air Force - 19

Preparing activity:
Air Force - 85

(Project 5910-1176-4)

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T03:00Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

C 16061

DD FORM 1 JAN 72 1426

REPLACES EDITION OF 1 JAN 68 WHICH MAY BE USED

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T03:00Z
Check the source to verify that this is the current version before use.

FOLD

POSTAGE AND FEES PAID

U.S.MAIL

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

AFLC/MMGE/Standardization
WPAFB OH 45433

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T03:00Z
Check the source to verify that this is the current version before use.